

range

```
In [1]: range(5)
```

```
Out[1]: range(0, 5)
```

```
In [3]: for i in range(5):  
        print(i)
```

```
0  
1  
2  
3  
4
```

```
In [4]: for i in range(0,5):  
        print(i)
```

```
0  
1  
2  
3  
4
```

```
In [8]: type(range(5))
```

```
Out[8]: range
```

```
In [9]: list(range(2,20))
```

```
Out[9]: [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19]
```

```
In [11]: list(range(2,20,5))    # start,stop,step
```

```
Out[11]: [2, 7, 12, 17]
```

```
In [12]: list(range(4,20,4,5))  #as kahi nste
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[12], line 1  
----> 1 list(range(4,20,4,5))  #as kahi nste  
TypeError: range expected at most 3 arguments, got 4
```

dict { : }

```
In [ ]: {keys:values}
```

```
In [13]: # not indexing, we take keys
```

```
In [67]: dict={}  
dict
```

Out[67]: {}

```
In [15]: d={'a':1,'b':2,'c':3}
         d
```

Out[15]: {'a': 1, 'b': 2, 'c': 3}

```
In [68]: type(dict)
```

Out[68]: dict

```
In [16]: d.keys()
```

Out[16]: dict_keys(['a', 'b', 'c'])

```
In [17]: d.values()
```

Out[17]: dict_values([1, 2, 3])

```
In [18]: d.items()
```

Out[18]: dict_items([('a', 1), ('b', 2), ('c', 3)])

```
In [19]: d[0]      # not indexing
```

```
-----
KeyError                                Traceback (most recent call last)
Cell In[19], line 1
----> 1 d[0]      # not indexing

KeyError: 0
```

```
In [21]: d['a']
```

Out[21]: 1

```
In [23]: d['c']
```

Out[23]: 3

```
In [26]: d['f'] = 5
         d      # add new
```

Out[26]: {'a': 1, 'b': 2, 'c': 3, 'f': 5}

```
In [27]: d[2+4j]=True
         d      # add new
```

Out[27]: {'a': 1, 'b': 2, 'c': 3, 'f': 5, (2+4j): True}

```
In [33]: keys=['a','b','c','d']      # from are unordered
         values=[10,20,30]
         dict=d.fromkeys(keys,values)
         dict
```

Out[33]: {'a': [10, 20, 30], 'b': [10, 20, 30], 'c': [10, 20, 30], 'd': [10, 20, 30]}

```
In [38]: values.append(50)
dict
```

```
Out[38]: {'a': [10, 20, 30, 50],
          'b': [10, 20, 30, 50],
          'c': [10, 20, 30, 50],
          'd': [10, 20, 30, 50]}
```

```
In [39]: d={10,20}
d
```

```
Out[39]: {10, 20}
```

```
In [41]: d.add(50)
d
```

```
Out[41]: {10, 20, 50}
```

```
In [43]: fs=frozenset(d)           # as it is,no change
fs                                             # we can't change
```

```
Out[43]: frozenset({10, 20, 50})
```

```
In [44]: type(fs)
```

```
Out[44]: frozenset
```

```
In [52]: d1={'a':1,'b':2,'c':{'A':5,'B':6},'d':[3,'t',8]} # dict in dict,list
d1
```

```
Out[52]: {'a': 1, 'b': 2, 'c': {'A': 5, 'B': 6}, 'd': [3, 't', 8]}
```

```
In [53]: d1[3] # no index
```

```
-----
KeyError                                Traceback (most recent call last)
Cell In[53], line 1
----> 1 d1[3]

KeyError: 3
```

```
In [57]: d1
```

```
Out[57]: {'a': 1, 'b': 2, 'c': {'A': 5, 'B': 6}, 'd': [3, 't', 8]}
```

```
In [58]: d1['c'] # only key
```

```
Out[58]: {'A': 5, 'B': 6}
```

```
In [61]: d1.get('a') # key-a
```

```
Out[61]: 1
```

```
In [62]: d1.get('e') # ans= none, key-e not available
```

```
In [69]: d={'a':1,'b':2,'c':3}
         d
```

```
Out[69]: {'a': 1, 'b': 2, 'c': 3}
```

```
In [72]: d2={'d':4,'e':5}
         d.update(d2)
```

```
In [73]: d
```

```
Out[73]: {'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}
```

```
In [78]: del[d['b']]
         d
```

```
Out[78]: {'a': 1, 'c': 3, 'd': 4, 'e': 5}
```

```
In [79]: d
```

```
Out[79]: {'a': 1, 'c': 3, 'd': 4, 'e': 5}
```

```
In [82]: d.pop('c')
```

```
-----
KeyError                                Traceback (most recent call last)
Cell In[82], line 1
----> 1 d.pop( )

KeyError: 'c'
```

```
In [83]: d
```

```
Out[83]: {'a': 1, 'd': 4, 'e': 5}
```

```
In [84]: d.popitem()           # random remove
```

```
Out[84]: ('e', 5)
```

```
In [85]: d4={'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}
         d4
```

```
Out[85]: {'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}
```

```
In [87]: d5=d4
         d5
```

```
Out[87]: {'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}
```

```
In [89]: print(id(d5))           # same id
         print(id(d4))
```

```
2632471092672
2632471092672
```

```
In [90]: print(d4)
         print(d5)
```

```
{'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}
{'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}
```

```
In [92]: d6=d5.copy()
d6
```

```
Out[92]: {'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}
```

```
In [93]: print(id(d5))
print(id(d6))    # diff id bcoz of copy

2632471092672
2632471272384
```

```
In [94]: d5
```

```
Out[94]: {'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}
```

```
In [97]: d7=d6.clear
```

```
In [98]: d7
```

```
Out[98]: <function dict.clear()>
```

```
In [99]: d4
```

```
Out[99]: {'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}
```

```
In [100... for i in d4:
            print(i)
```

```
a
b
c
d
e
```

```
In [101... for i in d4:
            print(i,':',d4[i])
```

```
a : 1
b : 2
c : 3
d : 4
e : 5
```

dict membership

```
In [102... d4
```

```
Out[102... {'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}
```

```
In [103... 'a' in d4
```

```
Out[103... True
```

```
In [104... 'f' in d4
```

Out[104... False

In [105... `10 not in d4`

Out[105... True

all /any

In [106... `d4`

Out[106... `{'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}`

In [107... `all(d4)`

Out[107... True

In [108... `any(d4)`

Out[108... True

In [111... `d={'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5, 'f': False}`
`d`

Out[111... `{'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5, 'f': False}`

In [112... `all(d)`

Out[112... True

In [113... `any(d)`

Out[113... True

In [119... `d1=d={'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5, 'f': False, 0: 10}`
`d1`

Out[119... `{'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5, 'f': False, 0: 10}`

In [120... `all(d1)`

Out[120... False

In [121... `any(d1)`

Out[121... True

In []: